

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12409541
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Hale; Noelle et al.

Oscillating multi-tool

Abstract

A power tool includes a housing, a motor that is disposed within the housing and defining a first axis, and a tool holder driven to oscillate about a second axis by the motor. The second axis is perpendicular to the first axis. The power tool also includes a vibration dampening assembly with a counterweight that is configured to reciprocate along a third axis that is perpendicular to both the first axis and the second axis in response to oscillation of the tool holder about the second axis.

Inventors: Hale; Noelle (San Diego, CA), Altoonisi; Husam M. (Tabuk, SA), Tarr; Ryan (Louisville, KY), Sanabria; Alec (Germantown, WI)

Applicant: MILWAUKEE ELECTRIC TOOL CORPORATION (Brookfield, WI)

Family ID: 84140867

Assignee: MILWAUKEE ELECTRIC TOOL CORPORATION (Brookfield, WI)

Appl. No.: 17/749765

Filed: May 20, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220388138 A1	Dec. 08, 2022

Related U.S. Application Data

us-provisional-application US 63191565 20210521

Publication Classification

Int. Cl.: B25D17/24 (20060101); B24B23/04 (20060101); B24B41/00 (20060101); B25F5/00 (20060101)

U.S. Cl.:

CPC **B25D17/24** (20130101); **B24B23/04** (20130101); **B24B41/007** (20130101); **B25F5/006** (20130101); B25D2217/0092 (20130101)

Field of Classification Search

CPC: B25F (5/006); B25D (17/24); B24B (23/04); B24B (41/007); B25B (21/02)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7938196	12/2010	Fischer	173/210	B25D 17/24
2010/0236806	12/2009	Heilig	173/217	B25F 5/006
2010/0307783	12/2009	Baumann	173/162.2	B25B 21/02
2020/0149609	12/2019	Zhang	N/A	B25F 5/025
2023/0150085	12/2022	Mizutani	451/6	B24B 23/005

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
207874150	12/2017	CN	N/A
3500405	12/2019	EP	N/A
2011000335	12/2010	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion for Application No. PCT/US2022/030283 dated Sep. 15, 2022 (10 pages). cited by applicant

Primary Examiner: Leeds; Daniel Jeremy

Attorney, Agent or Firm: Michael Best & Friedrich LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/191,565 filed on May 21, 2021, the entire content of which is incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present disclosure relates to power tools, and more particularly to oscillating multi-tools.

BACKGROUND OF THE INVENTION

(2) The present invention relates to power tools driven by an electric motor. Power tools utilize the rotation of a motor to provide useful torque for operations such as cutting, sanding, grinding, removing material, drilling, driving fasteners, and the like. One example of a power tool is an oscillating multi-tool. Oscillating multi-tools can be utilized with various accessories, such as blades and sanding or grinding pad attachments, for performing different functions.

SUMMARY OF THE INVENTION

(3) The present invention provides, in one aspect, a power tool including a housing, a motor that is

disposed within the housing and defining a first axis, and a tool holder driven to oscillate about a second axis by the motor. The second axis is perpendicular to the first axis. The power tool also includes a vibration dampening assembly with a counterweight that is configured to reciprocate along a third axis that is perpendicular to both the first axis and the second axis in response to oscillation of the tool holder about the second axis.

(4) In another aspect, the vibration dampening assembly further includes a first spring adjacent a first end of the counterweight and a second spring adjacent an opposite, second end of the counterweight. The first and second springs bias the counterweight toward a neutral position.

(5) In another aspect, the vibration dampening assembly further includes a case in which the counterweight is located.

(6) In a further aspect, the case is internally located within the housing or externally located of the housing.

(7) In another aspect, the case includes a ventilation port configured to communicate an interior of the case in which the counterweight is located with the atmosphere outside the case.

(8) The invention provides, in another aspect, a power tool including a housing, a motor disposed within the housing and defining a first axis, and a tool holder driven to oscillate about a second axis by the motor. The second axis being perpendicular to the first axis. The power tool also includes a drive mechanism configured to convert torque from the motor to an oscillating movement of a tool accessory received by the tool holder and a vibration dampening assembly including a counterweight configured to reciprocate along a third axis perpendicular to both the first axis and the second axis in response to oscillation of the tool holder about the second axis.

(9) The invention provides, in another aspect, a power tool including a housing, a motor disposed within the housing and defining a first axis, and a tool holder driven to oscillate about a second axis by the motor. The second axis being perpendicular to the first axis. The power tool also includes a vibration dampening assembly having a case supported by the housing and a counterweight positioned within the case. The counterweight is configured to reciprocate along a third axis perpendicular to both the first axis and the second axis in response to oscillation of the tool holder about the second axis. The vibration dampening assembly also includes a first spring arranged between the case and the counterweight and a second spring arranged between the case and the counterweight. The first spring biases the counterweight in a first direction and the second spring biases the counterweight in a second direction opposite the first direction. The first and second springs bias the counterweight toward a neutral position within the case.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a top perspective view of an oscillating multi-tool in accordance with an embodiment of the invention.

(2) FIG. 2 is a cross-sectional view of a portion of the multi-tool shown in FIG. 1.

(3) FIG. 3 is a perspective view of a vibration dampening assembly for use with the multi-tool of FIG. 1.

(4) FIG. 4 is an exploded view of the vibration dampening assembly of FIG. 3.

(5) FIG. 5 is a cross-sectional view of the vibration dampening module of FIG. 3.

(6) FIG. 6 is a perspective view of a case of another embodiment of the vibration dampening assembly of FIG. 3.

(7) FIG. 7 is a perspective view of another embodiment of a vibration dampening assembly for use with the multi-tool of FIG. 1.

(8) FIG. 8 is an exploded view of the vibration dampening assembly of FIG. 7.

(9) FIG. 9 is a cross-sectional view of the vibration dampening assembly of FIG. 7.

(10) Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

DETAILED DESCRIPTION

(11) FIGS. 1 and 2 illustrate a power tool, such as an oscillating multi-tool **10**, including a housing **12**, a motor **14**, a drive mechanism **16** that converts torque received from the motor **14** to an oscillating movement of a tool accessory **18** about an axis B, and a power source, such as a battery pack **22**, for powering the motor **14**. In some embodiments, the multi-tool **10** may be powered by an AC power source by way of a cord (not shown).

(12) With reference to FIG. 1, the housing **12** includes two clamshell halves **24a**, **24b** that are coupled together to enclose the motor **14** and the drive mechanism **16**. When connected, the clamshell halves **24a**, **24b** define a handle portion **26** and a battery support portion **28** of the housing **12** to which the battery pack **22** is attached. The handle portion **26** is configured to be grasped by a user during operation of the power tool **10**. An actuator **30** is located on the handle portion **26** of the housing **12** for switching the motor **14** between an “on” state and an “off” state. In some embodiments, a separate actuator may be employed for changing the rotational speed of the motor **14**. In other embodiments, the actuator **30** may additionally be operable to vary the rotational speed of the motor **14** in addition to switching the motor **14** between the “on” and “off” states.

(13) The motor **14** and the drive mechanism **16** are positioned within the housing **12** in front of the handle portion **26**. In the illustrated embodiment, the drive mechanism **16** is positioned within a gear case **32** attached to the housing **12** (FIG. 2). The motor **14** includes a drive shaft **34**, which provides a continuous torque input to the drive mechanism **16** when the motor **14** is in the “on” state. The drive mechanism **16** converts the continuous torque input from the drive shaft **34** into oscillating rotational motion of the tool accessory **18** about the axis B.

(14) The tool accessory **18** is coupled to an output shaft, or spindle **36**, of the drive mechanism **16**. In the illustrated embodiment, the spindle **36** defines the axis B, which is perpendicular to a longitudinal axis A defined by the housing **12** and the motor **14** (FIG. 2). When in the “on” state, the motor **14** drives the drive mechanism **16** to oscillate the spindle **36** and the tool accessory **18** about the axis B. In the illustrated embodiment, the tool accessory **18** is a cutting blade. However, in other embodiments, the tool accessory **18** may be a different type of blade such as a scraper blade, a circular blade, a semi-circular blade, etc., or a different type of element such as a sanding pad, a grinding element, etc.

(15) With continued reference to FIG. 2, the multi-tool **10** includes a clamping mechanism **20** for securing the tool accessory **18** to the spindle **36**. In the illustrated embodiment, the clamping mechanism **20** does not require the use of tools to attach, remove, and exchange tool accessories. The clamping mechanism **20** includes a plunger **38**, a spring **40**, a tool holder **42** coupled for co-rotation with the spindle **36**, and a fastener **44**, which will be described in greater detail below. A lever **46** is pivotable by a user to operate the clamping mechanism **20**.

(16) The lever **46** is pivotable about a pin **48**, which defines a pivot axis C, between a clamping position (FIG. 2) and a release position (not shown). In the clamping position, the tool accessory **18** is secured, or clamped, to the tool holder **42**. In the release position, the fastener **44** may be unthreaded from the plunger **38** to remove and/or exchange the tool accessory **18**. The lever **46** includes a cam **50** for displacing the plunger **38**. In the clamping position, the cam **50** does not engage the plunger **38**. In the release position, the cam **50** engages the plunger **38** to displace the plunger **38** against the bias of the spring **40**. A user may rotate the lever **46** about the pivot axis C, causing the cam **50** to engage and displace the plunger **38**, which releases the clamping force applied to the tool accessory **18** and allows the fastener **44** and the tool accessory **18** to be removed. A user may then exchange the tool accessory **18** and thread the fastener **44** into the plunger **38** so

that the tool accessory **18** reengages the tool holder **42**. Once the fastener **44** is repositioned, a user can rotate the lever **46** back to the clamping position, permitting the spring **40** to rebound and displace the plunger **38** and the connected fastener **44** toward the pivot axis C, which applies a clamping force between the tool holder **42** and the tool accessory **18**.

(17) FIGS. 3-5 illustrate a vibration dampening assembly **54** for use with the multi-tool **10**. The vibration dampening assembly **54** attenuates vibration created by the multi-tool **10** during operation. In some embodiments, the vibration dampening assembly **54** may be coupled to the housing **12** with a hose clamp, fasteners, or an adhesive. In other embodiments, the vibration dampening assembly **54** may be integrated with the housing **12**. As such, the vibration dampening assembly **54** may be coupled to the exterior of the housing **12**. In further embodiments, the vibration dampening assembly **54** may be disposed within the interior of the housing **12**. For example, the vibration dampening assembly **54** may be supported within the gear case **32**.

(18) As shown in FIGS. 4 and 5, the vibration dampening assembly **54** includes a case **58**, a counterweight **62** disposed within the case **58**, two end caps **66a**, **66b**, and two compression springs **70a**, **70b** positioned on opposite sides of the counterweight **62**. The case **58** is generally cylindrical and includes a first end **74** and a second end **78** opposite the first end **74**. The case **58** defines a bore **82** that extends between the first and second ends **74**, **78**. The case **58** also includes a bracket **86** that may be used to mount the vibration dampening assembly **54** to the housing **12**. Alternatively, in some embodiments, the case **58** may be integrally formed with the clamshell halves **24a**, **24b** or omitted entirely with the counterweight **62** being suspended within the housing **12**. In some embodiments, the case **58**, end caps **66a**, **66b**, and/or the counterweight **62** may be produced using an additive manufacturing process, such as 3D printing, and may be made from plastic, metal, or other suitable materials.

(19) With continued reference to FIG. 4, the counterweight **62** is also generally cylindrical and defines a first end **90** and a second end **94** opposite the first end **90**. The first end **90** of the counterweight **62** defines a first spring seat **98** to receive an end of the spring **70a** and the second end **94** defines a second spring seat **102** to receive an end of the other spring **70b**. A first end cap **66a** is coupled to the first end **74** of the case **58** and a second end cap **66b** is coupled to the second end **78** of the case **58**. As such, the first and second end caps **66a**, **66b** enclose the counterweight **62** and springs **70** within the bore **82** of the case **58**. Each end cap **66a**, **66b** includes a spring seat **106** that receives an end of one of the springs **70a**, **70b**. In some embodiments, one of the end caps **66a**, **66b** may be integrated with the case **58**.

(20) With reference to FIG. 5, the first spring **70a** is positioned between the spring seat **106** of the first end cap **66a** and the first spring seat **98** of the counterweight **62**. The second spring **70b** is positioned between the spring seat **106** of the second end cap **66b** and the second spring seat **102** of the counterweight **62**. The counterweight **62** reciprocates along an axis D, which is perpendicular to both of the axes A, B. In addition, the axis D is parallel with the pivot axis C.

(21) The first spring **70a** biases the counterweight **62** in a first direction, whereas the second spring **70b** biases the counterweight **62** in a second direction that is opposite the first direction. The first and second springs **70a**, **70b** have identical stiffnesses; therefore, the counterweight **62** is biased towards a neutral position between the end caps **66a**, **66b** when the motor **14** is in the “off” state and the tool accessory **18** is not oscillating. The first and second springs **70a**, **70b** suspend the counterweight **62** within the bore **82** so that the counterweight **62** can freely oscillate within the bore **82** without frictional contact with the interior of the case **58**. In addition, the counterweight **62** includes a smaller diameter than the bore **82** of the case **58** to inhibit the counterweight **62** from contacting the case **58**. The mass of the counterweight **62** and the stiffness of the springs **70a**, **70b** can be selected (i.e., tuned) to substantially dampen the vibration created by the oscillating movement of the drive mechanism **16** and the tool accessory **18** when the motor **14** is in the “on” state. In further embodiments, the vibration dampening assembly **54** may include more than two springs.

(22) During operation of the multi-tool **10**, in response to the drive mechanism **16** and the tool accessory **18** being oscillated by the motor **14**, vibration is created and translated to the operator through the housing **12** and handle portion **26**. However, the vibration dampening assembly **54** attenuates this vibration. Specifically, the drive mechanism **16** converts a continuous torque input from the motor **14** to oscillating rotation of the drive mechanism **16** and tool accessory **18** about the axis B. As torque is applied to the spindle **36**, the tool holder **42**, and the tool accessory **18** to incrementally rotate them about the axis B in a first rotational direction, a reaction torque is applied to the housing **12** in an opposite, second rotational direction about axis B. The spindle **36**, the tool holder **42**, and the tool accessory **18** then stop rotating in the first rotational direction, and their rotation is reversed to the second rotational direction. As the spindle **36**, the tool holder **42**, and the tool accessory **18** are rotated in the second rotational direction, a reaction torque is applied to the housing **12** in the first rotational direction about axis B. The oscillating reaction torque applied to the housing **12** creates the vibration felt by the user.

(23) The vibration dampening assembly **54** is offset from the axis B, with the reciprocating axis D of the counterweight **62** oriented transverse to the axis B, to create equal and opposite moments about the axis B to counteract the reversing reaction torque applied to the housing **12**. As such, the counterweight **62** reciprocates along the axis D out of phase with the oscillation of the drive mechanism **16** and the tool accessory **18** to counteract the reversing reaction torque. The springs **70a**, **70b** of the vibration dampening assembly **54** continually bias the counterweight **62** toward a neutral position. The reciprocating movement of the counterweight **62** reduces the magnitude of the vibration transmitted through the housing **12** and handle portion **26** to the user as the spindle **36**, the tool holder **42**, and the tool accessory **18** oscillate.

(24) FIG. **6** illustrates a case **110** of a vibration dampening assembly in accordance with another embodiment. The case **110** is similar to the case **58** but includes a slot **114** that extends into the bore **82**. The slot **114** ventilates the interior of the case **110** to prevent accumulated air pressure from affecting reciprocation of the counterweight **62**.

(25) FIGS. **7-9** illustrate a vibration dampening assembly **210**, in accordance with another embodiment of the invention, for use with the multi-tool **10** of FIG. **1**. The vibration dampening assembly **210** is similar to the vibration dampening assembly **54** with like features being represented with like references numerals. The vibration dampening assembly **210** includes a case **214**, a counterweight **62**, and first and second springs **70a**, **70b** (FIGS. **8** and **9**).

(26) With continued reference to FIGS. **8** and **9**, the case **214** includes an end cap **216** and a flange **218**. The flange **218** includes an indentation **222** that defines a spring seat **226** for the second spring **70b**. The end cap **216** includes an interior **230** (FIG. **9**) having a first cavity portion **234** and a second cavity portion **238** that has a smaller diameter than the first cavity portion **234**. The second cavity portion **238** defines a spring seat **242** for the first spring **70a**. Both the end cap **216** and the flange **218** include a plurality of bosses **246** extending from respective outer surfaces. A fastener **250** may be inserted through apertures in the bosses **246** to secure the end cap **216** to the flange **218**. In the illustrated embodiment, the fasteners **250** include a threaded bolt and a nut. In other embodiments, different types of fasteners may be used.

(27) With reference to FIG. **9**, when assembled, the case **214** defines an enclosed chamber **254** in which the counterweight **62** and the springs **70a**, **70b** are located. The first spring **70a** is positioned between the spring seat **242** of the flange **218** and the first spring seat **98** of the counterweight **62**. The second spring **70b** is positioned between the spring seat **226** of the end cap **216** and the second spring seat **102** of the counterweight **62**. Similar to the vibration dampening assembly **54**, the first spring **70a** biases the counterweight **62** in a first direction, whereas the second spring **70b** biases the counterweight **62** in a second direction that is opposite the first direction. The first and second springs **70a**, **70b** have identical stiffnesses; therefore, the counterweight **62** is biased towards a neutral position between the spring seats **226**, **242** when the motor **14** is in the “off” state. The first and second springs **70a**, **70b** suspend the counterweight **62** within the second cavity portion **238** of

the case **214** so that the counterweight **62** can freely oscillate within the second cavity portion **238** without frictional contact with the interior of the case **214**. In addition, the counterweight **62** includes a smaller diameter than the second cavity portion **238** to inhibit the counterweight **62** from contacting the case **214**.

(28) During operation of the power tool **10**, in response to the tool accessory **18** being oscillated by the motor **14**, vibration throughout the power tool **10** is generated and translated to the operator through the housing **12** and handle portion **26**. However, the vibration dampening assembly **210** attenuates this vibration. Specifically, the drive mechanism **16** converts a continuous torque input from the motor **14** to oscillating rotation of the drive mechanism **16** and the tool accessory **18** about the axis B. As torque is applied to the spindle **36**, the tool holder **42**, and the tool accessory **18** to incrementally rotate them about the axis B in a first rotational direction, a reaction torque is applied to the housing **12** in an opposite, second rotational direction about axis B. The spindle **36**, the tool holder **42**, and the tool accessory **18** then stop rotating in the first rotational direction, and their rotation is reversed to the second rotational direction. As the spindle **36**, the tool holder **42**, and the tool accessory **18** are rotated in the second rotational direction, a reaction torque is applied to the housing **12** in the first rotational direction about axis B. The oscillating reaction torque applied to the housing **12** creates the vibration felt by the user.

(29) The vibration dampening assembly **210** is offset from the axis B, with the reciprocating axis D of the counterweight **62** oriented transverse to the axis B, to create equal and opposite moments about the axis B to counteract the reversing reaction torque applied to the housing **12**. As such, the counterweight **62** reciprocates along the axis D out of phase with the oscillation of the drive mechanism **16** and the tool accessory **18** to counteract the reversing reaction torque. The springs **70a**, **70b** of the vibration dampening assembly **210** continually bias the counterweight **62** toward a neutral position. The reciprocating movement of the counterweight **62** reduces the magnitude of the vibration transmitted through the housing **12** and handle portion **26** to the user as the spindle **36**, the tool holder **42**, and the tool accessory **18** oscillate.

(30) Various features and advantages are set forth in the following claims.

Claims

1. A power tool comprising: a housing; a motor disposed within the housing and defining a first axis; a tool holder driven to oscillate about a second axis by the motor, the second axis being perpendicular to the first axis; a drive mechanism configured to convert continuous torque from the motor to continuous oscillating movement of a tool accessory received by the tool holder; and a vibration dampening assembly disposed in an interior of the housing, the vibration dampening assembly including a counterweight configured to reciprocate linearly along a third axis perpendicular to both the first axis and the second axis in response to oscillation of the tool holder about the second axis, the third axis being positioned above the first axis defined by the motor.
2. The power tool of claim 1, wherein the vibration dampening assembly further includes, a first spring adjacent a first end of the counterweight, and a second spring adjacent an opposite, second end of the counterweight, and wherein the first and second springs are coaxial with the third axis and bias the counterweight toward a neutral position.
3. The power tool of claim 2, wherein the first end of the counterweight defines a first spring seat to receive the first spring and the second end of the counterweight defines a second spring seat to receive the second spring.
4. The power tool of claim 1, wherein the vibration dampening assembly further includes a case in which the counterweight is positioned.
5. The power tool of claim 4, wherein the case includes a ventilation port configured to communicate an interior of the case in which the counterweight is located with the atmosphere outside the case.

6. The power tool of claim 4, wherein the counterweight is enclosed within the case.
 7. The power tool of claim 4, wherein the case includes a bracket configured to mount the vibration dampening assembly to the interior of the housing.
 8. The power tool of claim 1, wherein the counterweight is produced using an additive manufacturing process.
 9. The power tool of claim 1, wherein the counterweight is cylindrical.
 10. The power tool of claim 1, wherein the housing defines a battery support portion configured to receive a battery pack to power the motor.
 11. The power tool of claim 10, wherein the battery support portion is positioned at a first end of the housing and the drive mechanism is positioned at a second end opposite the first end.
 12. The power tool of claim 1, wherein the housing defines a handle portion configured to be grasped by a user, and wherein the motor and the drive mechanism are positioned within the housing in front of the handle portion.
 13. A power tool comprising: a housing; a motor disposed within the housing and defining a first axis; a tool holder driven to oscillate about a second axis by the motor, the second axis being perpendicular to the first axis; a drive mechanism configured to convert continuous torque from the motor to continuous oscillating movement of a tool accessory received by the tool holder; and a vibration dampening assembly disposed in an interior of the housing, the vibration dampening assembly including, a case supported by the housing, a counterweight positioned within the case, the counterweight configured to reciprocate linearly along a third axis perpendicular to both the first axis and the second axis in response to oscillation of the tool holder about the second axis, the third axis being positioned above the first axis defined by the motor, a first spring arranged between the case and the counterweight, the first spring biasing the counterweight in a first direction, and a second spring arranged between the case and the counterweight, the second spring biasing the counterweight in a second direction opposite the first direction; wherein the first and second springs are coaxial with the third axis and bias the counterweight toward a neutral position within the case.
 14. The power tool of claim 13, wherein a first end of the counterweight defines a first spring seat to receive the first spring and a second end of the counterweight opposite the first end defines a second spring seat to receive the second spring.
 15. The power tool of claim 13, wherein the vibration dampening assembly further includes a first end cap positioned adjacent a first side of the case and a second end cap positioned adjacent a second side of the case opposite the first side, the first and second end caps enclosing the counterweight within the case.
 16. The power tool of claim 13, wherein the counterweight is cylindrical.
 17. The power tool of claim 13, wherein the case includes a ventilation port configured to communicate an interior of the case in which the counterweight is located with the atmosphere outside the case.
 18. The power tool of claim 13, wherein the case includes a bracket configured to mount the vibration dampening assembly to the interior of the housing.
 19. The power tool of claim 1, wherein the counterweight is configured to reciprocate solely along the third axis.
-